

How Does Shape-Constrained Regularization Impact the Performance of EWMA, ARIMA, & GARCH Models in Multi-Horizon Volatility Forecasting for Financial Markets?

Nischal Adhikari

Abstract

Accurate forecasting of financial volatility is critical for risk management, derivative pricing, and portfolio optimization. Traditional models like Exponentially Weighted Moving Average (EWMA), AutoRegressive Integrated Moving Average (ARIMA), and Generalized Autoregressive Conditional Heteroskedasticity (GARCH) face challenges in high-noise environments, often overreacting to fluctuations or lacking alignment with market dynamics. This study explores the application of shape-constrained regularization, particularly monotonic constraints, to enhance model stability and interpretability. Using historical stock data from Apple (AAPL), Microsoft (MSFT), and Google (GOOG), we analyze the comparative performance of constrained and unconstrained models. Results demonstrate that while ARIMA and GARCH benefit modestly from constraints, EWMA exhibits limitations. Shape-constrained regularization shows promise for stabilizing forecasts in high-risk scenarios but introduces slight biases in low-complexity models.

1. Introduction

Financial volatility forecasting is central to assessing market risk and understanding asset price dynamics. It informs decisions in portfolio allocation, pricing of derivatives, and financial stress testing. Traditional models like EWMA, ARIMA, and GARCH provide frameworks for volatility forecasting but are often sensitive to short-term noise or overfitting, particularly during periods of market instability.

This study investigates whether shape-constrained regularization, which enforces non-decreasing constraints on forecasts, can address these challenges. By applying these constraints, the goal is to align model outputs with theoretical market behaviors, ensuring smooth and interpretable volatility trajectories.

The primary objective is to evaluate the performance of constrained versus unconstrained models across multiple metrics—Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and visual alignment with historical trends. This analysis focuses on a multi-horizon forecasting approach (5 days) using stock data from AAPL, MSFT, and GOOG.

2. Models and Methods

2.1 Volatility Forecasting Models

1. Exponentially Weighted Moving Average (EWMA)

EWMA is a simple, computationally efficient model defined as:

$$\sigma_t^2 = \lambda \cdot \sigma_{t-1}^2 + (1 - \lambda) \cdot r_{t-1}^2$$

Here, λ controls the weight given to past versus recent observations. While EWMA reacts quickly to new data, this often leads to overfitting in volatile markets.

2. ARIMA (AutoRegressive Integrated Moving Average)

ARIMA captures trends and autocorrelation in time series data, expressed as:

$$\Phi(B)(1 - B)^d y_t = \Theta(B)\epsilon_t$$

ARIMA models are suitable for stationary data and offer flexibility through parameters p , d , and q . However, they struggle with non-linear patterns inherent in financial markets.

3. GARCH (Generalized Autoregressive Conditional Heteroskedasticity)

GARCH extends ARIMA by modeling time-varying volatility using past variances and

returns:
$$\sigma_t^2 = \omega + \sum_{i=1}^p \alpha_i r_{t-i}^2 + \sum_{j=1}^q \beta_j \sigma_{t-j}^2$$

It effectively captures volatility clustering but is computationally intensive.

2.2 Shape-Constrained Regularization

Shape-constrained regularization introduces monotonicity constraints to ensure that forecasts are

non-decreasing:
$$\sigma_{t+1} \geq \sigma_t \quad \forall t$$

Implemented via convex optimization (CVXPY), this method minimizes the squared deviation between original and constrained forecasts, preserving smoothness while avoiding overreaction to short-term fluctuations.

3. Data and Exploratory Analysis

3.1 Data Source

Historical stock data for AAPL, MSFT, and GOOG were obtained, including daily closing prices and returns over a multi-year period.

3.2 Exploratory Data Analysis (EDA)

EDA revealed significant market volatility during major financial events, with rolling 30-day standard deviations highlighting trends. Moving averages were applied to identify long-term price trajectories. Similar patterns were seen across all three companies.

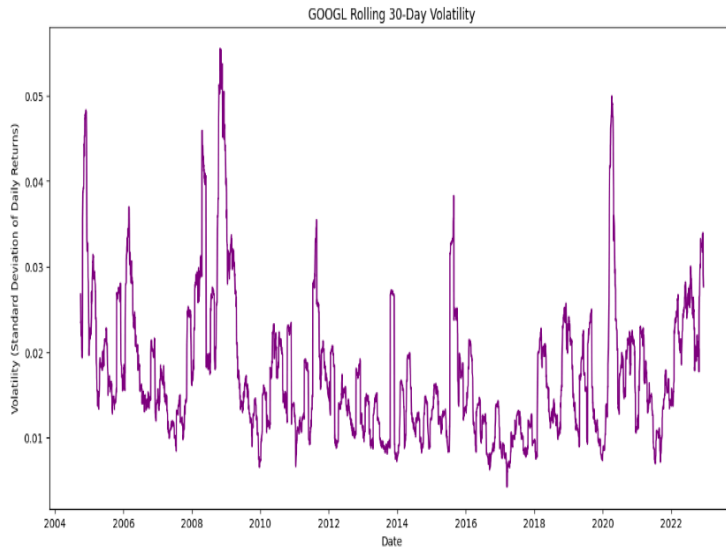


Figure 1 (Visual): Rolling standard deviation of daily returns for GOOG.

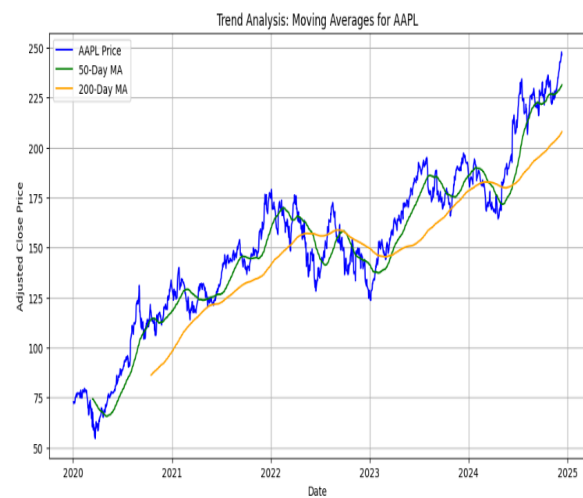
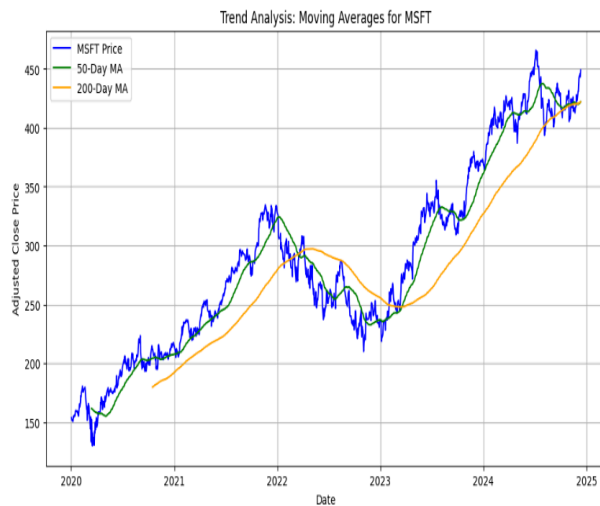


Figure 2 & Figure 3: Moving averages for MSFT & AAPL.

4. Methodology

Hyperparameters for each model were optimized using cross-validation:

- EWMA: λ optimized for minimal RMSE.
- ARIMA: p , d , & q determined via AIC minimization.
- GARCH: p & q selected to balance computational efficiency and accuracy.

Shape constraints were applied to the forecast outputs, and performance was evaluated over a 5-day horizon using RMSE & MAE. Visual alignment of constrained & unconstrained forecasts with historical volatility trends was also assessed.

5. Results

5.1 Error Comparison

Model	RMSE (Constrained)	RMSE (Unconstrained)	MAE (Constrained)	MAE (Unconstrained)
EWMA	0.1388	0.1269	0.1388	0.1265
ARIMA	0.0313	0.0313	0.0312	0.0312
GARCH	0.0217	0.0216	0.0216	0.0215

This table compares the RMSE and MAE values for constrained and unconstrained forecasts across EWMA, ARIMA, and GARCH models. It highlights that shape-constrained regularization slightly increases error for all models, with the most significant impact on the simpler EWMA model.

5.2 Visual Comparison

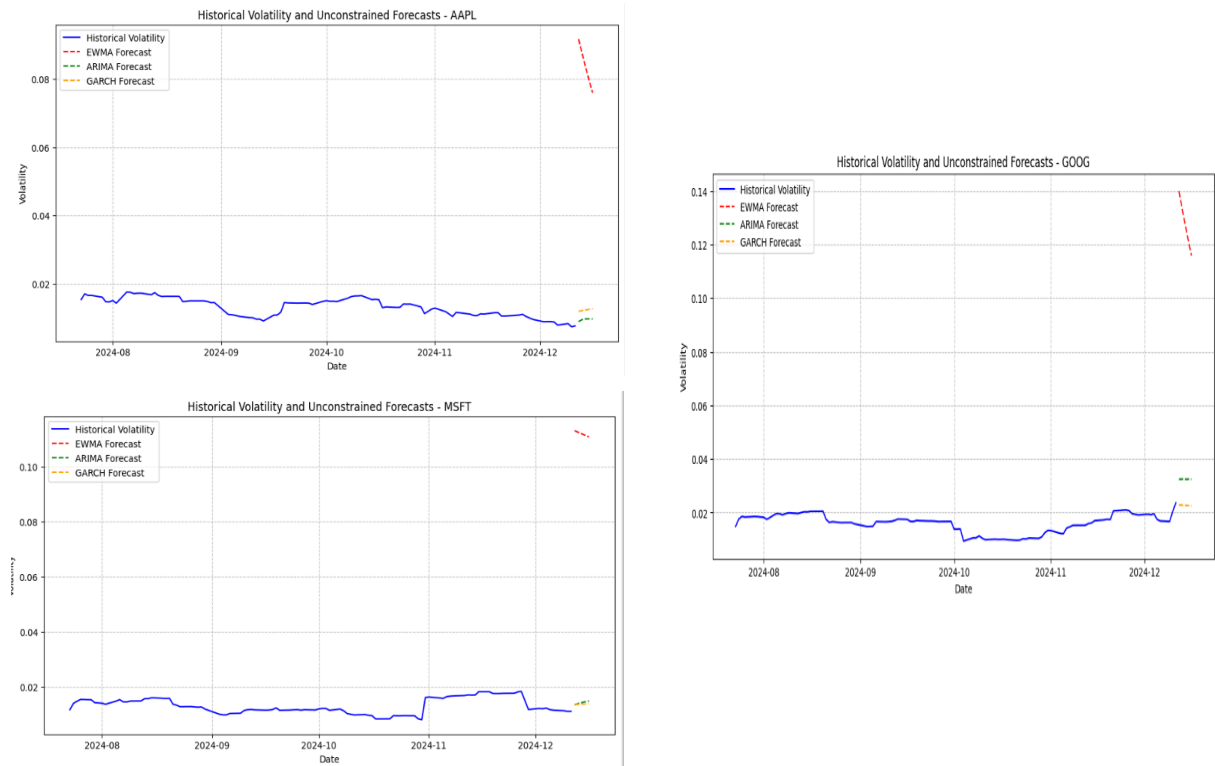


Figure 3, 4, 5: Historical volatility and forecasts for AAPL, MSFT, and GOOG (with and without constraints). Key observation: GARCH and ARIMA closely align with historical trends, while EWMA overreacts to recent changes.

5.3 Challenges

- Computational overhead for GARCH parameter tuning.
- Difficulty in achieving convergence during shape-constrained optimization.
- Sensitivity of EWMA to structural modifications, leading to higher errors.

6. Discussion

Shape-constrained regularization improves forecast stability by aligning predictions with theoretical expectations. However, the benefits are model-dependent. GARCH and ARIMA are robust even without constraints, while EWMA's simplistic structure limits its adaptability. Constraints introduce slight biases, highlighting the trade-offs between stability and accuracy.

7. Conclusion

Shape-constrained regularization offers a promising approach to enhance financial volatility forecasting, particularly under high-risk scenarios. However, its utility varies by model complexity, with GARCH and ARIMA showing modest improvements and EWMA facing challenges. Future research should explore alternative constraints, such as smoothness penalties, and test models under stress conditions for broader applicability.

8. Future Directions

1. **Stress Testing:** Evaluate models during volatile periods (e.g., financial crises).
2. **Alternative Constraints:** Experiment with penalties for smoother transitions rather than strict monotonicity.
3. **Exogenous Variables:** Incorporate macroeconomic indicators into ARIMA and GARCH models to enhance predictive power.

References

- fatimamansoor28. (2024, April 23). Performing EDA on stock market data. Kaggle.
<https://www.kaggle.com/code/fatimamansoor28/performing-eda-on-stock-market-data>
- Ho, S. L., & Xie, M. (1999, June 28). *The use of Arima models for reliability forecasting and analysis*. Computers & Industrial Engineering.
<https://www.sciencedirect.com/science/article/abs/pii/S0360835298000667>
- Mooney, P. (2023, January 16). Stock market data (NASDAQ, NYSE, S&P500). Kaggle.
<https://www.kaggle.com/datasets/paultimothymooney/stock-market-data>
- Nur, A. R., Jaya, A. K., & Siswanto, S. (2023). Comparative analysis of Ridge, lasso, and elastic net regularization approaches in handling multicollinearity for infant mortality data in South Sulawesi. *Jurnal Matematika, Statistika Dan Komputasi*, 20(2), 311–319.
<https://doi.org/10.20956/j.v20i2.31632>
- Pérez-Hernández, F., Arévalo-de-Pablos, A., & Camacho-Miñano, M.-M. (2024, December 9). *A hybrid model integrating artificial neural network with multiple GARCH-type models and EWMA for performing the optimal volatility forecasting of Market Risk Factors*. *Expert Systems with Applications*. <https://www.sciencedirect.com/science/article/pii/S0957417423033985>
- Y. Liang, A. Thavaneswaran, Z. Zhu, R. K. Thulasiram and M. E. Hoque, "Data-Driven Adaptive Regularized Risk Forecasting," 2020 IEEE 44th Annual Computers, Software, and Applications Conference (COMPSAC), Madrid, Spain, 2020, pp. 1296-1301, doi: 10.1109/COMPSAC48688.2020.00-77. keywords: {Predictive models;Reactive power;Forecasting;Correlation;Computational modeling;Adaptation models;Standards;Regularized Adaptive Forecasts, Data-Driven Volatility Model, EWMA, VaR, ES, Model Risk Ratio, Elastic Net, Probabilistic Forecasts},